

Watch-DevKit-HW

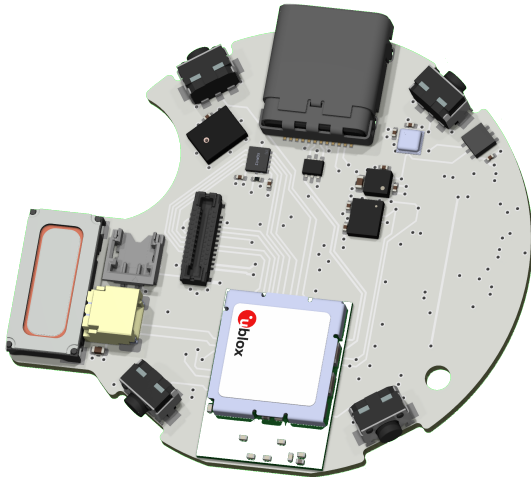
2026-02-21

Variant: RELEASED

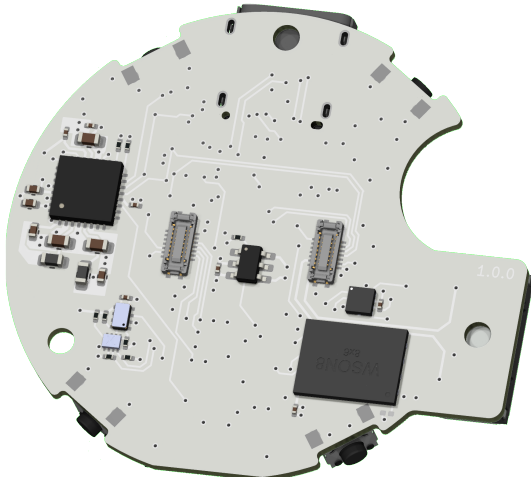
Rev 1.0.0

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TOP VIEW



BOTTOM VIEW



NOTES

Comment

Not fitted components are marked as **X**

DRAFT - Very early stage of schematic, ignore details.

PRELIMINARY - Close to final schematic.

CHECKED - There shouldn't be any mistakes. Contact the engineer if you find any.

RELEASED - A board with this schematic has been sent to production.

Date: 21-Feb-2026

DESIGN CONSIDERATIONS

DESIGN NOTE:
Example text for
informational design
notes.

DESIGN NOTE:
Example text for
debug notes.

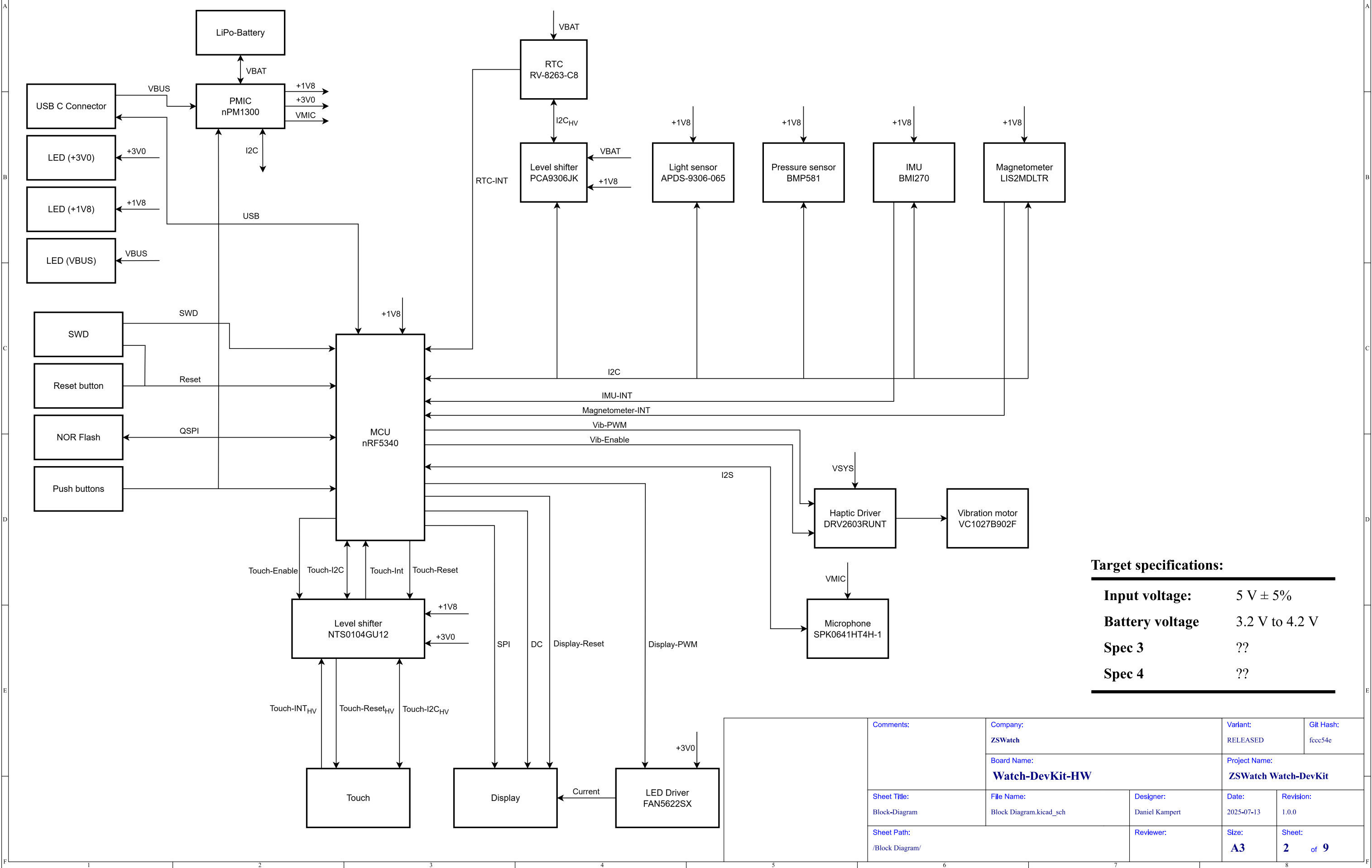
DESIGN NOTE:
Example text for
cautionary design
notes.

DESIGN NOTE:
Example text for
critical design
notes.

LAYOUT NOTE:
Example text for
critical layout
guidelines.

	Comments:	Company: ZSWatch	Variant: RELEASED	Git Hash: fccc54e
	Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Title: ZSWatch-Watch	File Name: ZSWatch-Watch.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13
	Sheet Path: /		Reviewer:	Revision: 1.0.0
			Size: A3	Sheet: 1 of 9

[2] Block-Diagram



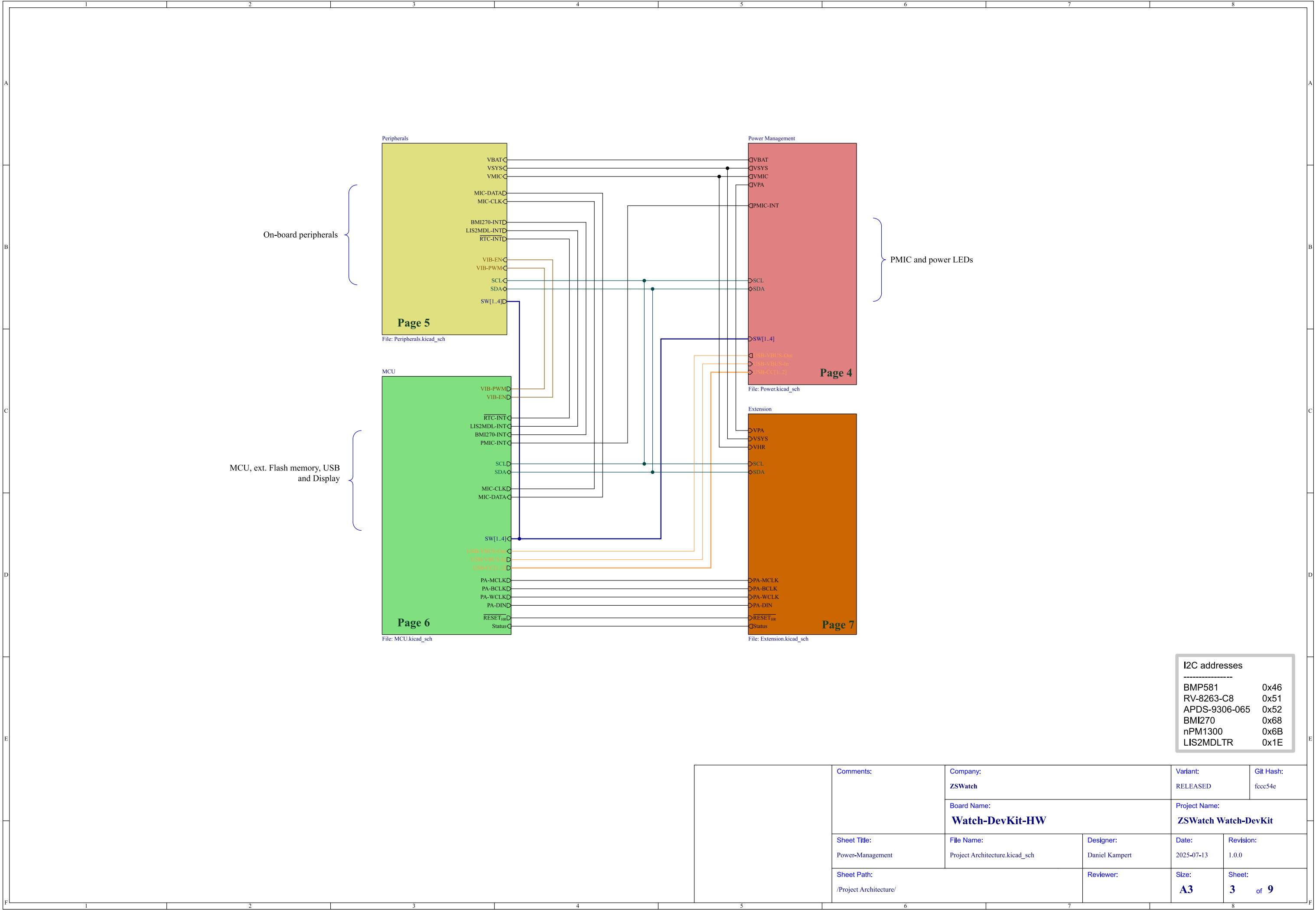
Target specifications:	
Input voltage:	5 V ± 5%
Battery voltage	3.2 V to 4.2 V
Spec 3	??
Spec 4	??

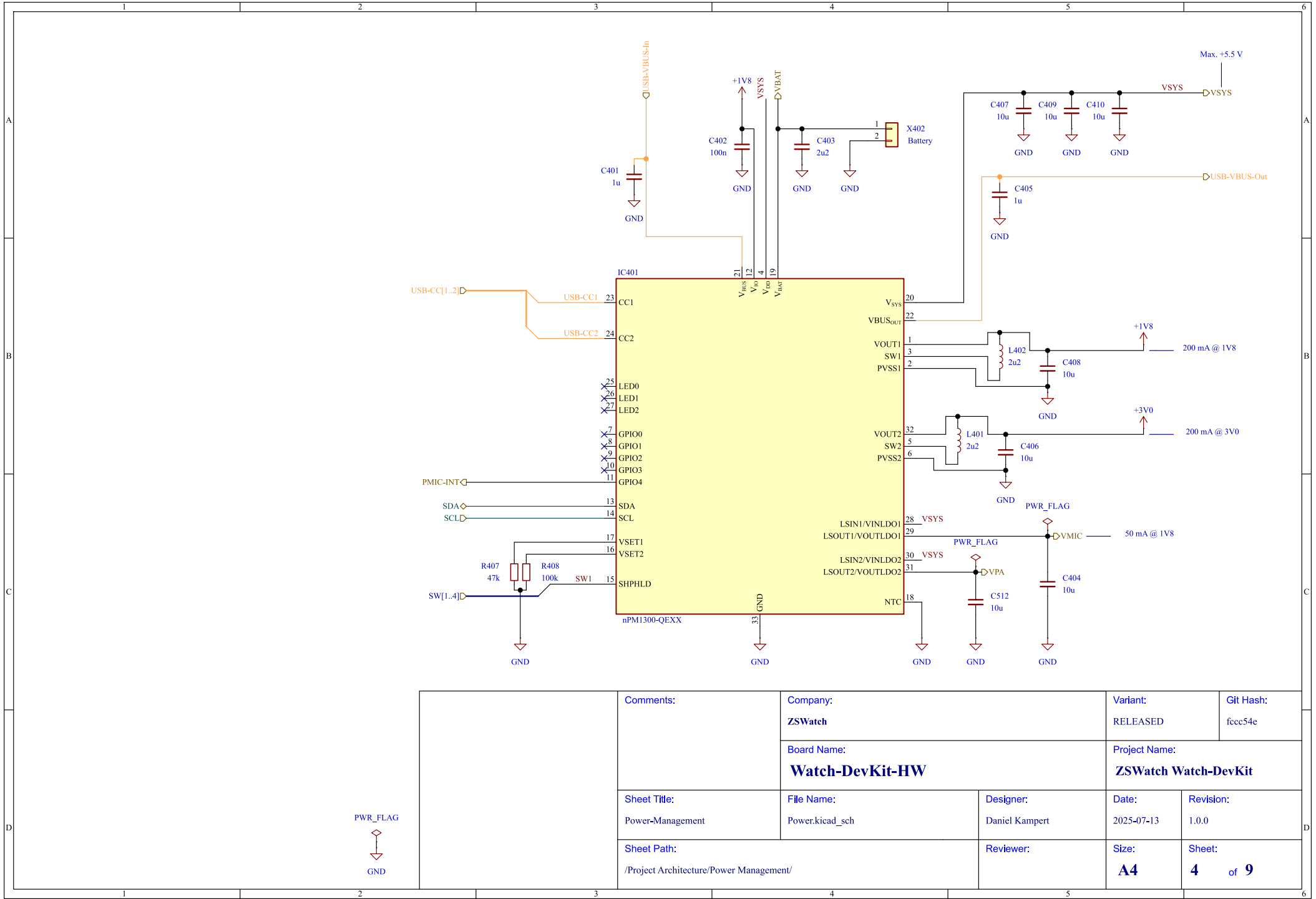
Comments:	Company: ZSWatch		Variant: RELEASED	Git Hash: fcc54e
	Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Title: Block-Diagram	File Name: Block Diagram.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13
Sheet Path: /Block Diagram/			Reviewer:	Size: A3
				Sheet: 2 of 9

	1	2	3	4	5	6	7	8
A								
B								
C								
D								
E								
F								

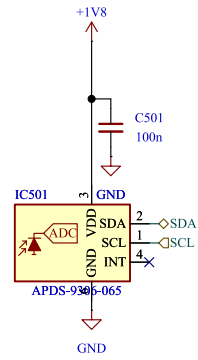
Target specifications:	
Input voltage:	?? - ?? V
Spec 2	??
Spec 3	??
Spec 4	??

	Comments:	Company: ZSWatch		Variant: RELEASED	Git Hash: fccc54e
		Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Title:	File Name: Power - Topology.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.0.0
	Sheet Path: /Power - Topology/		Reviewer:	Size: A3	Sheet: 2 of 9

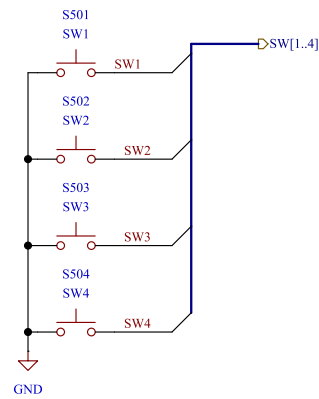




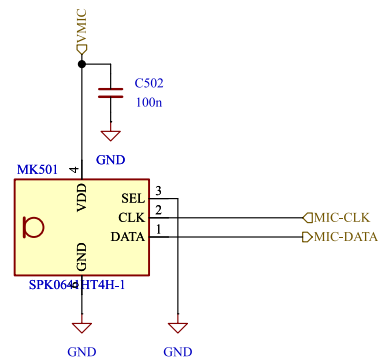
Light sensor



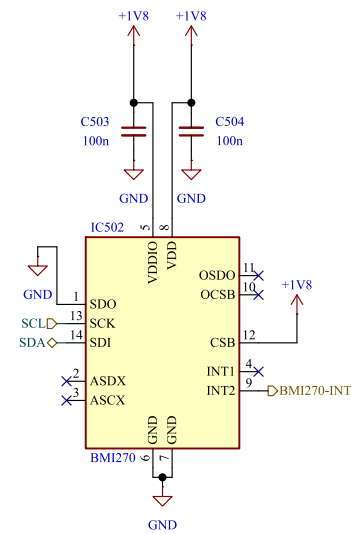
Buttons



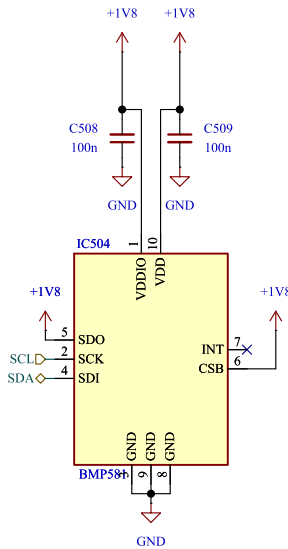
Microphone



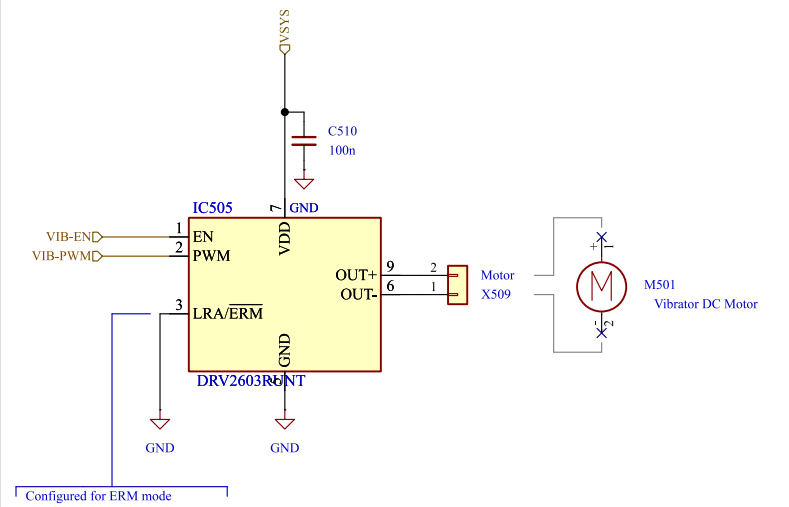
IMU with step counter and gesture detection



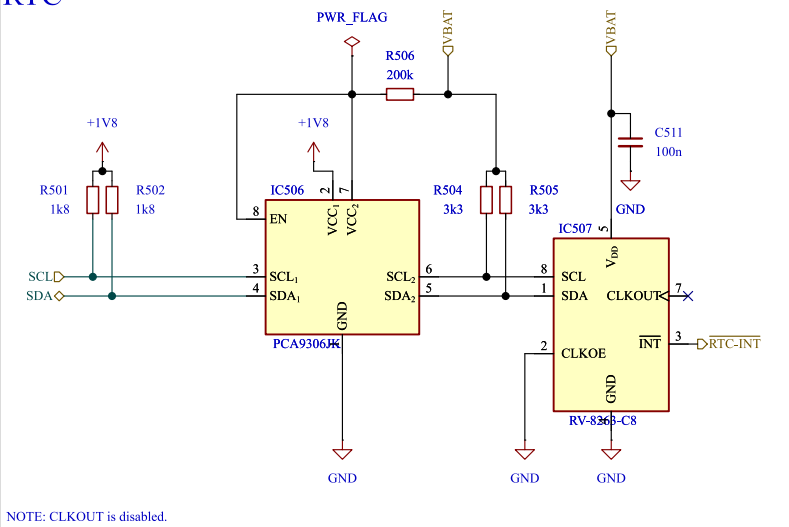
High accuracy pressure sensor



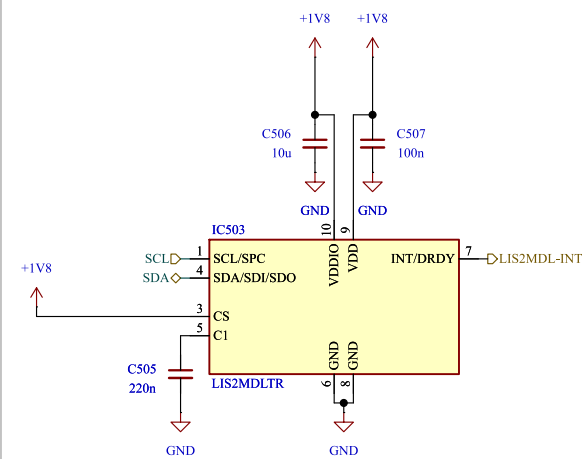
Vibration motor



RTC

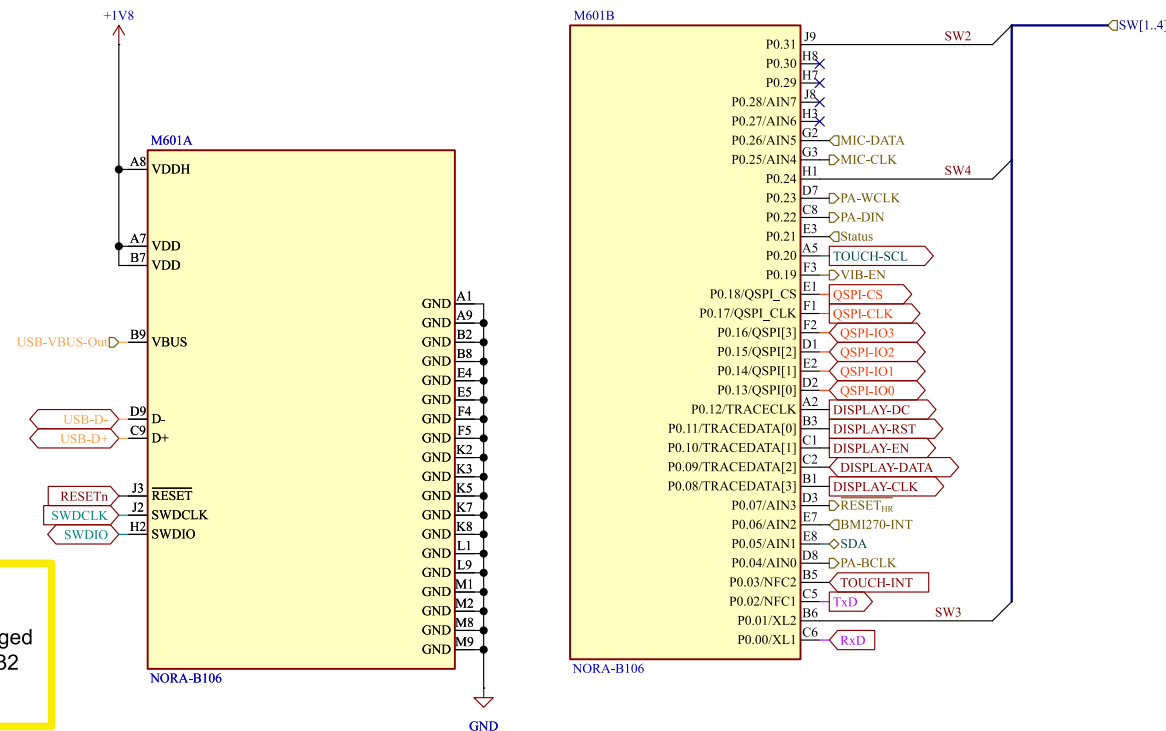


Magnetometer



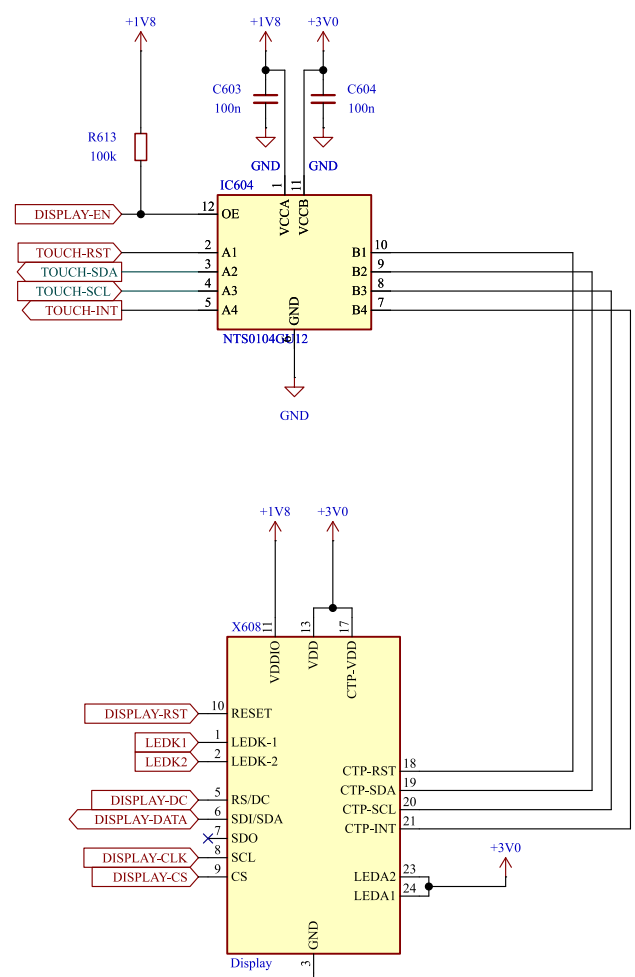
	Comments:	Company:	Variant:	Git Hash:
		ZSWatch	RELEASED	fccce54e
		Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit
Sheet Title:	File Name:	Designer:	Date:	Revision:
Peripherals	Peripherals.kicad_sch	Daniel Kampert	2025-07-13	1.0.0
Sheet Path:		Reviewer:	Size:	Sheet:
/Project Architecture/Peripherals/			A3	5 of 9

Microcontroller
(NORA-B106/nRF5340)



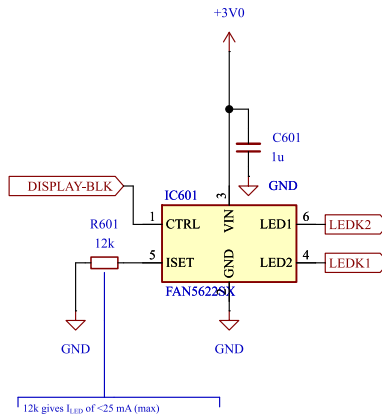
DESIGN NOTE:
The SPI lines for the display can't be changed and are required for 32 MHz operation.

Display



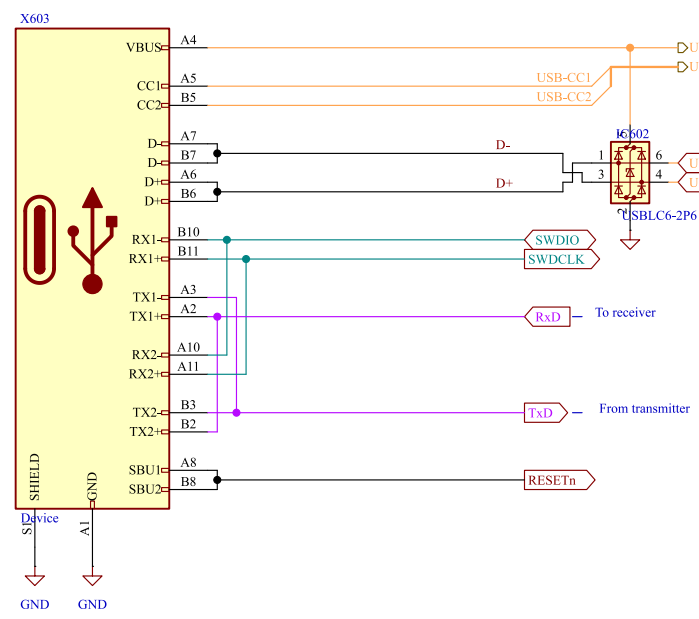
DESIGN NOTE:
NTS0104GU12 has internal pullups.

Display backlight

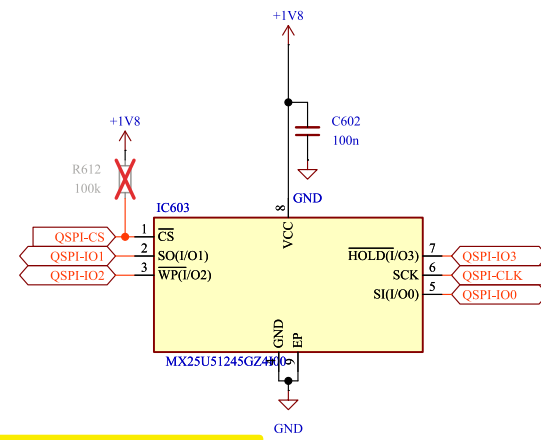


*12k gives I_{LED} of ~25 mA (max)

USB

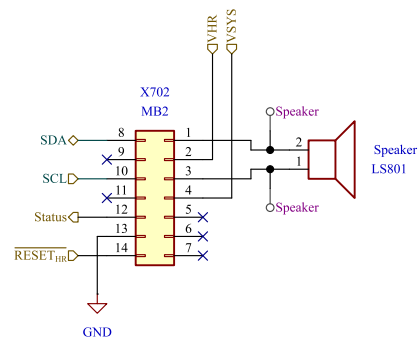


External Flash



DESIGN NOTE:
Make sure that the Flash Memory supports 4k sector erase commands (limited by QSPI driver)!

Comments:	Company: ZSWatch		Variant: RELEASED		Git Hash: fcc54e	
	Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit			
Sheet Title: MCU	File Name: MCU.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.0.0		
Sheet Path: /Project Architecture/MCU/		Reviewer:	Size: A3	Sheet: 6 of 9		



	Comments:	Company: ZSWatch		Variant: RELEASED	Git Hash: fccc54e	
		Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit		
	Sheet Title: Extension	File Name: Extension.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.0.0	
	Sheet Path: /Project Architecture/Extension/		Reviewer:	Size: A4	Sheet: 7 of 9	

[8] Revision-History

\${RELEASE_TITLE_1.0.0}
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	Comments:	Company: ZSWatch		Variant: RELEASED	Git Hash: fccc54e
		Board Name: Watch-DevKit-HW		Project Name: ZSWatch Watch-DevKit	
	Sheet Title: Revision-History	File Name: Revision History.kicad_sch	Designer: Daniel Kampert	Date: 2025-07-13	Revision: 1.0.0
	Sheet Path: /Revision History/		Reviewer:	Size: A4	Sheet: 8 of 9